Contents

1 Day 1. Classic binomial and multinomial identities

Planned entrance quiz :)

1. How many ways are to order 6 persons in a queue?
2. The box contains 50 balls numbered from 1 to 50. We draw 6 balls without replacement and write down the resulting sequence of numbers. How many different sequences are possible?
3. I have 5 yellow books and 7 red books. I want to place them on my bookshelf and I would like to place yellow books to the left of red books. How many book orders are possible?

Definition 1.1. Factorial of n, denoted by $n!$, is the number of ways to put n distinct objects in a queue.

Definition 1.2. Binomial coefficient $\binom{n}{k}$ is the number of ways to select k distinct objects from n distinct objects.

Definition 1.3. For $n = a + b + c$ the multinomial coefficient $\binom{n}{a,b,c}$ is the number of ways to split n distinct objects into different boxes, where the first box contains a objects, the second box contains b objects, the third box contains c objects.

One may create multinomial coefficient for any number of boxes :)

Exercise 1.1. Is $\binom{n}{k}$ equal to $\binom{n}{n-k}$?

Is $\binom{n}{k}$ equal to $\binom{n}{k,n-k}$?

Exercise 1.2. Write the product

$$\binom{100}{20} \cdot \binom{80}{50} \cdot \binom{30}{10}$$

as one multinomial coefficient.

Exercise 1.3. Write

$$\binom{n}{a, b, c, d}$$

using binomial coefficients.

Write

$$\binom{n}{a, b, c, d}$$

using factorials of n, a, b, c and d.

Idea 1. Solution strategies for combinatorial identities:

1. Double counting or story proof.
2. Visual identity (rare gem!).
3. Evaluation of a polynomial at a special point. Differentiation, integration.
4. Combine old identities into new.
5. Guess the answer and verify by induction.
6. Dark art!

Pascal's triangle and rule

Exercise 1.4. Recall the Pascal arithmetic triangle. Draw two versions: triangular and rectangular.

Exercise 1.5. Pascal's rule. Simplify

$$\binom{n}{k} + \binom{n}{k+1}.$$

$$\binom{n}{a-1, b, c} + \binom{n}{a, b-1, c} + \binom{n}{a, b, c-1}.$$

$$\binom{n}{a-1, b, c, d} + \binom{n}{a, b-1, c, d} + \binom{n}{a, b, c-1, d} + \binom{n}{a, b, c, d-1}.$$

Exercise 1.6. Simplify

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

Recall the Pascal's rule and simplify

$$\binom{2n}{0} + \binom{2n}{2} + \binom{2n}{4} + \cdots + \binom{2n}{2n}.$$

Simplify

$$\binom{2n-1}{0} + \binom{2n-1}{2} + \binom{2n-1}{4} + \cdots + \binom{2n-1}{2n-2}.$$

Exercise 1.7. Simplify

$$\sum_{a+b+c=2026} \binom{2026}{a, b, c}.$$

$$\sum_{a+b+c+d=2027} \binom{2027}{a, b, c, d}.$$

$$\sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}.$$

$$\sum_{a+b \leq n} \binom{n}{a} \cdot \binom{n-a}{b}.$$

Here a, b and c are assumed non-negative integers.

Hockey-stick identity

Exercise 1.8. Hockey-stick identity , Christmas stocking identity, boomerang identity.
Simplify

$$\sum_{n=a}^{b} \binom{n}{a}.$$

$$1 + 2 + 3 + \cdots + n.$$

$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n \cdot (n+1).$$

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \cdots + n \cdot (n+1) \cdot (n+2).$$

Exercise 1.9. Obtain closed form expressions for:

$$1^2 + 2^2 + \cdots + n^2,$$

$$1^3 + 2^3 + \cdots + n^3,$$

$$1^4 + 2^4 + \cdots + n^4.$$

Exercise 1.10. Hockey-sticks plus idea from linear algebra!

a) Decompose $n^2 + 2n + 5$ using $\binom{n}{0}$, $\binom{n}{1}$, $\binom{n}{2}$ as the basis.

b) Find the sum

$$\sum_{k=2}^{2026} k^2 + 2k + 5.$$

c) Using similar trick find the sums

$$\sum_{k=1}^{n} 5k^2 + 4k, \quad \sum_{k=1}^{n} k^3 + 2k^2.$$

Exercise 1.11. Let's start from some number $\binom{n}{k}$ inside the Pascal's triangle.
In one step we can go either in north-east or north-west direction.
Find the sum of all numbers we can reach.

Binomial or Taylor or MacLaurin expansion

Exercise 1.12. Write the Taylor expansion (binomial expansion) at $x = 0$ for

$$(1+x)^n, \quad (x+y)^n.$$

Exercise 1.13. By plugging some special points to the $(x+y)^n$ or somewhere else find the sums

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

$$\sum_{k=0}^{n} k\binom{n}{k}, \quad \sum_{k=0}^{n}(k-1)\binom{n}{k}, \quad \sum_{k=0}^{n}(-1)^k\binom{n}{k}, \quad \sum_{k=0}^{n}(-1)^{k-1}k\binom{n}{k}$$

Hint: differentiation may help you.

How that can be?

Exercise 1.14. Be brave, really brave!

a) Take a deep breath!

b) Calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/3}{k}, \quad \binom{-1/2}{n}.$$

Exercise 1.15. a) Simplify

$$\binom{-1/2}{3} + \binom{-1/2}{4}.$$

b) Draw the Pascal's triangle (the whole half-plane!) starting from $\binom{-1/2}{0}$.

c) Draw the Pascal's triangle (the whole half-plane!) starting from $\binom{1/3}{0}$.

Exercise 1.16. Write the Taylor expansion at $x = 0$ for

$$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

Exercise 1.17. Simplify

$$\binom{10}{0}\binom{-0.5}{7}+\binom{10}{1}\binom{-0.5}{6}+\binom{10}{2}\binom{-0.5}{5}+\cdots+\binom{10}{7}\binom{-0.5}{0}.$$

Hint: for a honest proof consider the binomial expansion of $(1+x)^{1}0$, $(1+x)^{-0.5}$, $(1+x)^{10-0.5}$.

Exercise 1.18. Homage to Newton.

a) Write the Taylor expansion at $x=0$ for $(1-x^2)^{1/2}$.

b) Visually (or analytically if you wish) find the integral

$$\int_{-1}^{1}(1-x^2)^{1/2}.$$

c) Calculate the sums

$$\binom{1/2}{0}-\frac{1}{3}\binom{1/2}{1}+\frac{1}{5}\binom{1/2}{2}+\ldots$$

$$\sum_{k=2}^{\infty}\frac{1}{2k+1}\frac{(2k-3)!!}{(2k)!!}$$

Home assignment: day 1 $\to$ day 4

1. Simplify

$$\sum_{a+b+c=100}\binom{100}{a}\cdot\binom{100-a}{b}.$$

2. Find explicit formula for the sum

$$1^3+2^3+\cdots+n^3.$$

   Hint: represent the general term k^3 as the sum of binomial coefficients and then use the hockey stick or Christmas stocking or boomerang!

3. Write the Taylor expansion at $x=0$ for

$$\frac{1}{1-x},\quad (1+x)^{-5},\quad (1+x)^{-1/2},\quad (1+3x)^{1/3},\quad (1-5x)^{1/2}.$$

4. Watch a Veritasium video «Newton changed the game», [Ver21].

   https://youtu.be/gMlf1ELvRzc?si=VASoVKQj3Pk-wstA, .

Demo for the quiz on day 2

1. [*] Using hockey stick decomposition find the sums

$$\sum_{k=1}^{n} 5k^2 + 4k.$$

2. [*] Take a deep breath and calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/3}{k}, \quad \binom{-1/2}{n}.$$

You are more than welcome to solve other exercises that we skipped. Starred exercises will be on the quiz next day.

2 Day 2. Stars, bars and Gauss summation trick

Day 2 welcome quiz!

1. Decompose the function $h(k) = 2k^2 + k$ using the binomial coefficient basis $\binom{k}{0}$, $\binom{k}{1}$ and $\binom{k}{2}$.

2. Let's play hockey! Find the sum

$$\sum_{k=2}^{1000} 3\binom{k}{0} - 2\binom{k}{1} + 3\binom{k}{2}.$$

Hint: do not forget to bring money back to the bank if you borrow!

3. Find the value

$$\binom{1/3}{2} + \binom{1/3}{3} + \binom{4/3}{2}.$$

Gauss summation trick

Exercise 2.1. Simplify

$$1 + 2 + 3 + \cdots + n;$$

$$1^2 + 2^2 + 3^2 + \cdots + n^2;$$

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n};$$

$$\binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \cdots + (n+1)\binom{n}{n};$$

$$\binom{n}{2} + 2\binom{n}{3} + 3\binom{n}{4} + \cdots + (n-1)\binom{n}{n};$$

$$\binom{n}{0} + 3\binom{n}{1} + 5\binom{n}{2} + \cdots + (2n+1)\binom{n}{n};$$

$$\binom{n}{0} - 2\binom{n}{1} + 3\binom{n}{2} - 4\binom{n}{3} \cdots + (-1)^n(n+1)\binom{n}{n};$$

$$3\binom{n}{1}+7\binom{n}{2}+11\binom{n}{3}+\cdots+(4n-1)\binom{n}{n};$$
$$\binom{n}{1}-2\binom{n}{2}+3\binom{n}{3}-4\binom{n}{4}+\cdots+(-1)^n n\binom{n}{n};$$

Stars and bars

Exercise 2.2. There are 20 chairs in a row and 7 students. Each student needs one chair to sit down. We do care only about which chairs are occupied.

a) In how many ways chairs can be occupied?

b) In how many ways chairs can be occupied if there should be at least one empty chair between occupied chairs?

c) In how many ways chairs can be occupied if there should be at least two empty chairs between occupied chairs?

Exercise 2.3. I have four bags of different colors and 20 identical marble stones.

a) In how many different ways can I put marbles into bags?

b) In how many different ways can I put marbles into bags if there should be no empty bags?

Theorem 2.1 (Stars and bars theorem)**.** The number of ways to distribute n identical objects into k distinguishable bags is equal to
$$\binom{n+k-1}{n}=\binom{n+k-1}{k-1}.$$
Hint: n is the number of stars and $(k-1)$ is the number of bars.

Exercise 2.4. There are 5 types of post-cards sold at the post-office.

a) In how many different ways can I buy 12 post-cards?

b) In how many different ways can I buy 12 post-cards if I would like to buy at least one post-card of every type?

Exercise 2.5. Consider the equation $x_1+x_2+x_3+x_4=2026$.

a) How many solutions does it have if x_1, x_2, x_3 and x_4 are non-negative integers?

b) How many solutions does it have if x_1, x_2, x_3 and x_4 are positive integers?

c) How many solutions does it have if all x_i are integers and $x_i \geq 2$ for all i?

d) How many solutions does it have if all x_i are integers and $x_i \geq i$ for all i?

Exercise 2.6. There are 10 standard indistinguishable dices.

a) How many different combinations of scores are possible?

b) How many different combinations of scores are possible if at least one dice shows 6?

c) How many different combinations of scores are possible if no dice shows 5 nor 6?

Exercise 2.7. Stars, bars, circles and probability!
Consider a well shuffled deck of 52 cards. I open the cards one by one.

a) How many cards on average I open to see the first Queen?

b) How many cards on average I open to see the third Ace?

c) What is expected number of cards between the third and the fourth King?

d) What is the probability that the third Ace will be the 20-th card opened?

e) Write combinatorial identities hidden in a), b) and c).

Visualize it

Exercise 2.8. a) Express as a binomial coefficient and visualize the sum:

$$1 + 2 + 3 + 4 + \cdots + n.$$

b) Visualize the Fuji-sum and simplify it!

$$1 + 2 + 3 + \cdots + (n-1) + n + (n-1) + \cdots + 3 + 2 + 1.$$

c) Recall the multiplication table for numbers from 1 to 10. Why the sum of all numbers is just

$$1^3 + 2^3 + 3^3 + \cdots + 10^3?$$

Simplify the sum!

Home assignment: day 2 $\to$ day 5

Deadline: 2026-07-03, 16:59.

1. There are 10 standard indistinguishable dices.

   a) How many different combinations of scores are possible?
   b) How many different combinations of scores are possible if at least one dice shows 6?
   c) How many different combinations of scores are possible if no dice shows 5 nor 6?

2. There are 5 types of post-cards sold at the post-office.

   a) In how many different ways can I buy 12 post-cards?
   b) In how many different ways can I buy 12 post-cards if I would like to buy at least one post-card of every type?

3. Using Gauss summation trick or otherwise find the sum:

$$\binom{n}{0} - 2\binom{n}{1} + 3\binom{n}{2} - 4\binom{n}{3} \cdots + (-1)^n (n+1)\binom{n}{n};$$

4. Recall the mother-identity

$$(1+x)^n = 1 + \binom{n}{1}x^1 + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n.$$

   (a) Find the first derivative of the mother-identity.
   (b) Find the value of the sum

$$\binom{n}{1} + 2 \cdot 7 \cdot \binom{n}{2} + 3 \cdot 7^2 \cdot \binom{n}{2} + \cdots + n \cdot 7^{n-1}\binom{n}{n}.$$

   (c) Find the second derivative of the mother-identity.
   (d) Find the value of the sum

$$2 \cdot 1\binom{n}{2} + 3 \cdot 2 \cdot 9 \cdot \binom{n}{3} + 4 \cdot 3 \cdot 9^2 \cdot \binom{n}{4} + 5 \cdot 4 \cdot 9^3 \cdot \binom{n}{5} + \cdots + n(n-1)9^{n-2}\binom{n}{n}.$$

Demo for the quiz on day 3

1. Using Gauss summation trick or otherwise find the sum

$$3\binom{n}{1} + 5\binom{n}{2} + 7\binom{n}{3} + \cdots + (2n+1)\binom{n}{n}.$$

2. Consider the equation $a + b + c = 100$ with integers $a \geq 3$, $b \geq 2$, $c \geq 5$.
   How many solutions are there?

3 Day 3. Recurrence relations

Exercise 3.1. Recurrence equations starter pack :)

a) Solve the recurence equation:

$$a_n = a_{n-1} + 3, \quad a_1 = 20.$$

b) Solve the recurence equation:

$$b_n = 3 \cdot b_{n-1}, \quad b_1 = 20.$$

c) Solve the recurence equation:

$$c_n = (n+3) \cdot c_{n-1}, \quad c_1 = 20.$$

d) Solve the recurence equation:

$$d_n = n + 3 + d_{n-1}, \quad d_1 = 20.$$

e) Solve the recurrence equation:

$$e_n = n^2 - 3n + 2 + e_{n-1}, \quad e_1 = 20.$$

f) Solve the recurrence equation:

$$f_n = (n^2 - 3n + 2)f_{n-1}, \quad f_1 = 20.$$

g) Solve the recurrence equation:

$$g_n = \frac{n+2}{n+1} g_{n-1}, \quad g_1 = 20.$$

h) Solve the recurrence equation:

$$h_n = \frac{1}{n^2 + n} + h_{n-1}, \quad h_1 = 20.$$

Exercise 3.2. Gauss trick with substraction!

a) Find the sum

$$S = 2 + 2 \cdot (1/3) + 2 \cdot (1/3)^2 + 2 \cdot (1/3)^3 + \ldots$$

b) Solve the recurrence equation

$$S_n = S_{n-1} + 3 \cdot (1/5)^n, \quad S_0 = 0.$$

c) Solve the recurrence equation

$$S_n = S_{n-1} + n \cdot (1/5)^n, \quad S_0 = 0.$$

Exercise 3.3. You draw n lines on a plane, they split the plane into some number of regions. Lines may or may not coincide.

a) What is the minimal number of regions a_n?

b) What is the maximal number of regions b_n?

Hint: start by finding b_1, b_2, then find the recurrence equation for b_n.

Idea 2. Consider the equation

$$x_t + \alpha_1 x_{t-1} + \alpha_2 x_{t-2} = \text{poly(t)}.$$

The solution can be always represented as a sum or product of some geometric series and polynomials.

Exercise 3.4. Unique root case.

a) Solve the recurence equation:

$$a_n - 5a_{n-1} + 6a_{n-2} = 0, \quad a_0 = 3, a_1 = 5.$$

b) Solve the recurence equation:

$$a_n - 7a_{n-1} + 12a_{n-2} = 6, \quad a_0 = 1, a_1 = 12.$$

c) Solve the recurence equation:

$$a_n - 8a_{n-1} + 15a_{n-2} = 8n - 22, \quad a_0 = 5, a_1 = 22.$$

Exercise 3.5. We start with one pair of young rabbits $F_1 = 1$. It takes one period of time for a pair of rabbits to mature. Every mature pair of rabbits in one period of time will give birth to a new young pair of rabbits.

a) Find F_2, F_3, F_4, F_5.

b) Deduce the recurrence relation for F_n.

c) Find the general formula for F_n.

d) Which term in the general formula can be ignored for the big n?

e) Guess and prove formulas for

$$\sum_{k=0}^{n} F_k, \quad \sum_{k=0}^{n} F_{2k}, \quad \sum_{k=1}^{n} F_{2k-1}, \quad F_{n+1}F_{n-1} - F_n^2.$$

f) Which of the following formulas are valid?

$$\begin{aligned} F_{m+n} &= F_m F_{n+1} + F_{m-1}F_n \\ F_n^2 + F_{n+1}^2 &= F_{2n+1} \\ F_{n+1}^2 - F_{n-1}^2 &= F_{2n} \\ \sum_{k=0}^{n} F_k^2 &= F_n F_{n+1}. \end{aligned}$$

Exercise 3.6. The mighty lag operator!

Let L be the lag operator and $\Delta = 1 - L$. Consider the sequences $x_n = 5^n$, $y_n = n \cdot 5^n$ and $z_n = 3^n$.

a) Find $(1 - 3L)x_n$, Δx_n, $(1 - 5L)x_n$.

b) Find $\Delta(1 - 3L)x_n$, $\Delta(1 - 5L)x_n$, $(1 - 5L)^2 x_n$.

c) Find $(1 - 3L)y_n$, Δy_n, $(1 - 5L)y_n$.

d) Find $\Delta(1 - 3L)y_n$, $\Delta(1 - 5L)y_n$, $(1 - 5L)^2 y_n$.

e) Find $(1 - 3L)(1 - 5L)x_n$ and $(1 - 3L)(1 - 5L)z_n$.

f) Rewrite the equation $a_t - 7a_{t-1} + 12a_{t-2} = 0$ using lag operator.

Warning! Remember that $(1 - 3L)$ or $(1 - 5L)$ is not a number! Lag operator is just a convinient notation to shift index!

Exercise 3.7. Double root case.

a) Solve the recurence equation:

$$a_n - 8a_{n-1} + 16a_{n-2} = 0, \quad a_0 = 3, a_1 = 0.$$

b) Solve the recurence equation:

$$a_n - 4a_{n-1} + 13a_{n-2} = 20, \quad a_0 = 6, a_1 = 4.$$

c) Solve the recurence equation:

$$q_n - 6q_{n-1} + 9q_{n-2} = 8n - 20, \quad q_0 = 2, a_1 = 12.$$

Home assignment: day 3 → day 6

Deadline: 2026-07-06, 14:59.

1. Solve the recurrence equation:
$$a_n = (n+1) + a_{n-1}, \quad a_1 = 100.$$
Hint: recall that $1+2+3+4+\cdots+n = \binom{n+1}{2} = \frac{n(n+1)}{2}$.

2. Solve the recurence equation:
$$a_n + 5a_{n-1} + 6a_{n-2} = 0, \quad a_0 = 3, a_1 = 5.$$

3. Solve the recurence equation:
$$c_n = 2 \cdot c_{n-1} + 3, \quad c_1 = 20.$$

4. Solve the recurence equation:
$$a_n - 7a_{n-1} + 12a_{n-2} = 6, \quad a_0 = 1, a_1 = 12.$$

5. Watch a Veritasium video «This equation will change how you see the world», [Ver20].

https://youtu.be/ovJcsL7vyrk?si=kh8wjM2e6dgLSXzj, .

Demo for the quiz on day 4

1. Solve the recurrence equation:
$$b_n = \frac{n+5}{n+4} b_{n-1}, \quad b_1 = 100.$$

2. Here L is the lag operator and $x_n = 6^n$. Find the values
$$(1-5L)x_n, \quad (1-6L)x_n, \quad (1-6L)(1-5L)x_n.$$

4 Day 4. More recurence equations

Exercise 4.1. Double root case.

a) Solve the recurence equation:
$$a_n - 8a_{n-1} + 16a_{n-2} = 0, \quad a_0 = 3, a_1 = 0.$$

b) Solve the recurence equation:
$$a_n - 4a_{n-1} + 4a_{n-2} = 20, \quad a_0 = 20, a_1 = 21.$$

c) Solve the recurence equation:
$$q_n - 6q_{n-1} + 9q_{n-2} = 8n - 20, \quad q_0 = 2, a_1 = 12.$$

Exercise 4.2. Unique complex root case.

a) Solve the recurence equation:

$$a_n - 4a_{n-1} + 13a_{n-2} = 0, \quad a_0 = 4, a_1 = 2.$$

b) Solve the recurence equation:

$$a_n - 4a_{n-1} + 13a_{n-2} = 20, \quad a_0 = 6, a_1 = 4.$$

c) Solve the recurence equation:

$$a_n + 9a_{n-2} = 10n - 8, \quad a_0 = 7, a_1 = 8.$$

Exercise 4.3. I enter a casino in Monte Carlo with 50 tokens. I play the roulette. I always bet one token on black. I will leave the casino when my fortune will be 70 tokens or 0 tokens.

In a typical Monte Carlo roulette there are 18 black pockets, 18 red pockets and zero. Let p_n be the probability that I will leave the casino as winner if I have n tokens now.

a) What are the initial conditions p_0 and p_{70}?

b) Build the recurrence equation for p_n.

c) Solve the recurrence equation and find p_{50}.

d) How the answers would change for Las Vegas roulette with zero and double zero pockets?

e) How the answers would change for an abstract fair casino with no zero nor double zero?

Let e_n be the expected number of spins before I leave the casino (as winner or as loser).

6. What are the initial conditions e_0 and e_{70}?
7. Build the recurrence equation for e_n.
8. Solve the recurrence equation and find e_{50}.
9. How the answers would change for Las Vegas roulette with zero and double zero pockets?
10. How the answers would change for an abstract fair casino with no zero nor double zero?

Exercise 4.4. I have a broken light switch button in my room. The switch button should turn the light on if the light is off and vice versa. On the first press the button works with probability $1/2$, on the second press — with probability $1/3$, on the third one — with probability $1/4$ and so on. Initially my room is dark.

What is the probability that my room will be dark after I press the button 100 times?

Home assignment: day 4 → day 7

Deadline: 2026-07-07, 14:59.

1. Solve the recurence equation:

$$a_n + 6a_{n-1} + 9a_{n-2} = 0, \quad a_0 = 5, a_1 = 10.$$

2. Solve the recurence equation:

$$c_n + 6c_{n-1} + 10c_{n-2} = 0, \quad c_0 = 0, c_1 = 10.$$

3. Solve the recurrence equation:

$$a_n = (n+5)a_{n-1}, \quad a_1 = 100.$$

   You may certainly use factorial in your answer!

4. I enter a casino in Las Vegas with 50 tokens. I play the roulette. I always bet one token on black. I will leave the casino when my fortune will be 70 tokens or 0 tokens.

   In a typical Las Vegas roulette there are 38 pockets: 18 black pockets, 18 red pockets, one zero and one double zero. Let p_n be the probability that I will leave the casino as winner if I have n tokens now.

   a) What are the initial conditions p_0 and p_{70}?

   b) Build the recurrence equation for p_n.

   c) Solve the recurrence equation for p_n.

   d) Evaluate p_{50} up to 2 digits.

Demo for the quiz on day 5

1. Consider the recurrence equation $y_n - 7y_{n-1} + 10y_{n-2} = 0$.

   Find all solutions of the form $y_n = \lambda^n$.

2. You know that $r_n = c \cdot 2^n + d \cdot n \cdot 2^n + 4$ is the solution of some recurrence equation with initial conditions $r_0 = 6$ and $r_1 = 10$.

   Find c and d.

5 Day 5. Systems of linear recurrence equations, examples from probability

Exercise 5.1. Now the systems!

a) Solve the recurrence equation:

$$\begin{cases} a_n = 3a_{n-1} + 2b_{n-1} - 4, \\ b_n = 2a_{n-1} + 3b_{n-1} - 4, \\ a_0 = 1, \quad b_0 = 0. \end{cases}$$

b) Solve the recurrence equation:

$$\begin{cases} a_n = 3a_{n-1} - 2b_{n-1} - 4, \\ b_n = -2a_{n-1} + 3b_{n-1} + 4, \\ a_0 = 1, \quad b_0 = 0. \end{cases}$$

Exercise 5.2. There are n passengers boarding an intercity bus and standing in queue. There are n places and each passenger has its own place on the ticket. The problem is that the first passenger in the queue is very nervous, he rushes in the bus and takes one place at random. Each next passenger takes his seat according to the ticket if the seat is free. And if his own seat is occupied each passenger takes a random free seat.

Let p_n be the probability that the last passenger in the queue will get his seat.

a) What is the initial conditions p_2?

b) Build the recurrence equation for p_n.

c) Guess the solution!

Exercise 5.3. I have 100 safes. Each safe can be closed without a key, but requires a specific unique key to be opened. I randomly distribute 100 keys into 100 safes, one key per safe. I leave 5 safes open and I close other safes. So I have access to 5 keys and I can use them to open other safes and inside these safes there are more keys...

Let's denote by p_{kn} the probability that I can open all n safes, given that in the beginning there are k safes open and $(n-k)$ safes closed.

a) What are the initial conditions p_{0n} and p_{nn}?

b) Build the recurrence equation for p_{kn}.

c) Guess the solution!

Let's denote by e_{kn} the expected number of open safes, given that in the beginning there are k safes open and $(n-k)$ safes closed.

4. What are the initial conditions e_{0n} and e_{nn}?

5. Build the recurrence equation for e_{kn}.

6. Guess the solution!

Exercise 5.4. You play a the following lottery with the casino. There is a hat on the table. A fair dice is thrown until the game ends.

- If the dice shows one then you get $+1$ dollar immediately and the game continues.
- If the dice shows two then $+2$ dollars are added to the hat and the game continues.
- If the dice shows three then $+3$ dollars are added to the hat and the game continues.
- If the dice shows four then the sum in the hat goes back to casino and the game continues.
- If the dice shows five then the sum in the hat goes back to the casino and the game ends.
- If the dice shows six then the game ends and you receive the content of the hat.

Let e_n be your total expected payoff if there are n dollars in the hat now.

1. Build the recurrence equation for e_n.
2. Find the solution.
3. What is the price of one dollar in the hat in real dollars? What is the fair price to participate in this game?

6 Day 6. Generating functions

Idea 3. Any sequence of numbers

$$(a_0, a_1, a_2, a_3, \dots)$$

can be written as a generating function

$$g(t) = a_0 + a_1 t^1 + a_2 t^2 + a_3 t^3 + \dots$$

Other variants of a generating function may be also useful

$$m(t) = a_0 + a_1 \exp(t) + a_2 \exp(2t) + a_3 \exp(3t) + \dots$$

or

$$e(t) = a_0 + a_1 t^1/1! + a_2 t^2/2! + a_3 t^3/3! + \dots$$

or even

$$f(t) = a_0 + a_1 \cos(t) + a_2 \cos(2t) + a_3 \cos(3t) + \dots$$

Definition 6.1. We will denote by $\operatorname{coef}(g(t), t^k)$ the coefficient a_k before the term t^k in the expansion

$$g(t) = a_0 + a_1 t^1 + a_2 t^2 + a_3 t^3 + \dots + a_k t^k + \dots$$

Exercise 6.1. Master the extraction of coefficients!

a) For the function $g(t) = (1+t)^{100}$ find $\operatorname{coef}(g, t^2)$, $\operatorname{coef}(g, t^{50})$.

b) For the function $g(t) = 1/(1-t)$ find $\operatorname{coef}(g, t^2)$, $\operatorname{coef}(g, t^{50})$, $\operatorname{coef}(g, t^{2026})$.

c) For the function $g(t) = \frac{1}{t(t+1)}$ find $\operatorname{coef}(g, t^k)$.

d) For the function $g(t) = \sqrt{1-t^2}$ find $\operatorname{coef}(g, t^{100})$ and $\operatorname{coef}(g, t^{101})$.

e) For the function $g(t) = (1 + x^5 + x^7)^{20}$ find $\operatorname{coef}(g, x^{17})$.

Exercise 6.2. Do you understand how coefficients are related to derivatives?

a) You know that $g(0) = 5$, $g'(0) = 7$, $g''(0) = 12$, $g'''(0) = 20$. Find $\operatorname{coef}(g, 1)$, $\operatorname{coef}(g, 1)$, $\operatorname{coef}(g, 2)$, $\operatorname{coef}(g, 3)$.

b) Write the relation between $\operatorname{coef}(g, t^k)$ and k-th derivative, $\frac{\partial^k}{\partial t^k} g(0)$.

c) For the function $g(t) = 1/(1-t)^p$ where $p \in \mathbb{N}$ find $\operatorname{coef}(g, t^2)$, $\operatorname{coef}(g, t^{50})$, $\operatorname{coef}(g, t^k)$.

Idea 4. How to discover the equation for the generating function?

- Represent the studied objects as vertices of a tree. Here a_n will be the number of objects with distance from the root equal to n.
- Encode objects as words, where a_n is the number of n-letter words. These words correspond to path from the root to the object.
- Start from recurrence equation for a_n and use sigma-boy approach.
- Start from recurrence equation for a_n and use two-way ticket approach.
- Apply first step approach to generating function in probability theory.

Exercise 6.3. Replacing the Taylor expansion by the closed formula!

a) Find the closed formula for
$$g(x) = 1 + x + x^2 + x^3 + x^4 + \ldots$$

b) Find the closed formula for
$$g(x) = 1 + 7x + 7^2x^2 + 7^3x^3 + 7^4x^4 + \ldots$$

c) Find a more compact formula for
$$g(x) = 1 + x + x^2 + x^3 + x^4 + \cdots + x^{2026}.$$

d) Find a more compact formula for
$$g(x) = 1 + \binom{10}{1}x + \binom{10}{2}x^2 + \binom{10}{3}x^3 + \cdots + \binom{10}{10}x^{10}.$$

e) Find the closed formula for
$$g(x) = 0 + x + 2x^2 + 3x^3 + 4x^4 + \ldots$$
Hint: one may use differentiation or n-letter words approach.

f) Find the closed formula for
$$g(x) = 1 \cdot 2x^2 + 2 \cdot 3x^3 + 3 \cdot 4x^4 + \ldots$$

g) Find the closed formula for
$$g(x) = \binom{2}{2}x^2 + \binom{3}{2}x^3 + \binom{4}{2}x^4 + \ldots$$

h) Find the closed formula for
$$g(x) = 0 + x + 2^2x^2 + 3^2x^3 + 4^2x^4 + \ldots$$
Hint: sum up some previous results

Exercise 6.4. Consider the rabbits story behind the Fibonacci numbers:

We start with one pair of young rabbits $F_1 = 1$. It takes one period of time for a pair of rabbits to mature. Every mature pair of rabbits in one period of time will give birth to a new young pair of rabbits.

For consistency with the recurrence equation assume that $F_0 = 0$.

a) Draw the tree where each rabbit alive at time n is a node that is n-vertices apart from the root.

b) Encode each rabbit alive at time n as n-letter word. Hint: two types of letters are enough.

c) Find the equation for the generating function.

d) Find the generating function.

e) Using the generating function find

$$\sum_{k=0}^{\infty} F_k/100^k, \quad \sum_{k=0}^{\infty} kF_k/100^k.$$

f) Represent 0.010102030508132134 55 approximately as a simple fraction.

g) Recover the formula for Fibonacci numbers using «sigma-boy» approach.

h) Recover the formula for Fibonacci numbers using «two-way ticked» approach.

Exercise 6.5. Consider the recurrence equation

$$a_n = 3a_{n-1} - 1, \quad a_0 = 1.$$

a) Find the equation for the generating function using «sigma-boy» method.

b) Find the closed form of the generating function from the equation.

c) Find the closed form of the generating function using «two-way ticket» method.

d) Find the formula for a_n and the value a_{1000}.

Exercise 6.6. Consider the recurrence equation

$$a_n = 3a_{n-1} + 2n - 1, \quad a_1 = 4.$$

a) Find a_0 such that the recurrence formula works.

b) Find the equation for the generating function using «sigma-boy» method.

c) Find the closed form of the generating function from the equation.

d) Find the closed form of the generating function using «two-way ticket» method.

e) Find the formula for a_n and the value a_{1000}.

Exercise 6.7. Convolution!

Consider two generating functions

$$A(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots, \quad B(x) = b_0 + b_1x + b_2x^2 + b_3x^3 + \ldots$$

and they product $C(x) = A(x) \cdot B(x)$.

a) Find c_0, c_1, c_2, c_3.

b) Find the general formula for c_k.

Exercise 6.8. Consider the recurrence equation

$$\binom{n+4}{4} = \sum_{0}^{k} a_k a_{n-k}, \quad a_0 = 1.$$

1. Find a_0, a_1, a_2 by bare hands.
2. Find the equation for the generating function $g(t)$.
3. Find the formula for a_n.

Home assignment: day 6 $\to$ day 9

1. Consider the recurrence equation
$$a_n = 3a_{n-1} + 5, \quad a_0 = 1.$$
   (a) Find equation for the generating function $g(t)$.
   (b) Obtain $g(t)$ in a closed form.
   (c) Extract coefficients (a_n) from $g(t)$.
2. Extract the sequence (a_n) from the generating functions
$$g(t) = \frac{1}{(1-t)^{50}}, \quad g(t) = \frac{1}{(1-3t)^{100}}.$$
3. Find the generating function of the following sequences
$$(1, 67, 67^2, 67^3, 67^4, 67^5, \ldots), \quad \left(\binom{5}{0}, \binom{6}{1}, \binom{7}{2}, \binom{8}{3}, \ldots\right).$$
   Hint: you don't need to solve equations, just recognize the patterns we have already seen.
4. You know that the generating function $g(t)$ of some sequence $(a_0, a_1, a_2, \ldots)$ satisfies the equation
$$g(t) = 4 - 9t + (5t - 6t^2)g(t).$$
   (a) Find $g(t)$.
   (b) By simply plugging some value of t inside $g(t)$ find the sum
$$\sum_{n=0}^{\infty} \frac{a_n}{100^n}.$$

Demo for the quiz on day 7

1. Extract the sequence (a_n) from the generating functions:
$$g(t) = \frac{1}{1-3t}, \quad g(t) = \frac{1}{1+7t}, \quad g(t) = \frac{t}{1-3t}.$$
2. Decompose the fraction into a sum of more simple fractions
$$g(t) = \frac{4-9t}{(1-2t)(1-3t)}.$$

7 Day 7. Moorrreee generating functions

Exercise 7.1. You can write any positive integer number on the sides of a dice.

a) Is it possible to create two different non-standard dices Y_1 and Y_2 such that the distribution of $Y_1 + Y_2$ is the same as the distribution of a sum of two fair dices $X_1 + X_2$?

b) Is it possible to create two identical non-standard dices Y_1 and Y_2 such that the distribution of $Y_1 + Y_2$ is the same as the distribution of a sum of two fair dices $X_1 + X_2$?

c) Is it possible to create two different non-standard dices Y_1 and Y_2 such that the distribution of $Y_1 + Y_2$ is uniform from 2 to 12?

d) Is it possible to create two identical non-standard dices Y_1 and Y_2 such that the distribution of $Y_1 + Y_2$ is uniform from 2 to 12?

e) Is it possible to create one non-standard dice Y_1 such that the distribution of sum with a standard second dice $Y_1 + X_2$ is uniform from 2 to 12?

Exercise 7.2.

8 Day 8. Happy day ;)

Midterm demo

1. You have 5 cups of different colors and 20 identical straws.

   (a) In how many ways you can place straws into cups?

   (b) In how many ways you can place straws into cups if at least 3 cups should be non-empty?

2. Simplify

$$\binom{5}{0}\binom{6}{4} + \binom{5}{1}\binom{6}{3} + \binom{5}{2}\binom{6}{2} + \binom{5}{3}\binom{6}{1} + \binom{5}{4}\binom{6}{0}.$$

3. Solve the recurrence equation

$$a_n = \frac{n(n+1)}{(n+2)(n+3)} a_{n-1}, \quad a_0 = 100.$$

   Hint: find a_1, a_2, a_3 and guess the pattern.

4. You play a the following lottery with the casino. There is a hat on the table. A fair dice is thrown until the game ends.

   - If the dice shows one then you get $+1$ dollar immediately and the game ends.
   - If the dice shows two then the sum in the hat is multiplied by 2 and the game continues.
   - If the dice shows three then $+3$ dollars are added to the hat and the game continues.
   - If the dice shows four then the sum in the hat goes back to casino and the game ends.
   - If the dice shows five then the sum in the hat goes back to the casino and the game continues.

- If the dice shows six then the game ends and you receive the content of the hat.

Let e_n be your total expected payoff if there are n dollars in the hat now.

Find the recurrence equation for e_n. You don't need to solve it.

5. Consider the recurrence equation

$$a_n - 4a_{n-1} + 5a_{n-2} = 0 \text{ for } n \geq 2, \quad a_0 = 5, a_1 = 2.$$

Find the generating function $g(t)$ for the sequence $(a_0, a_1, a_2, \ldots)$.

6. The sequence $(b_0, b_1, b_2, \ldots)$ has a generating function

$$g(t) = \frac{2}{1-3t} + \frac{5}{1-2t}.$$

Find b_0, b_1 and b_{100}.

Midterm exam

1. I take 10 cards from a well-shuffled standard deck of 52 cards.

   (a) What is the probability that I will get exactly 2 aces?

   (b) What is the probability that I will get 3 aces and 2 kings?

2. Simplify

$$\binom{5}{0}\binom{6}{2} + \binom{5}{1}\binom{6}{3} + \binom{5}{2}\binom{6}{4} + \binom{5}{3}\binom{6}{5} + \binom{5}{4}\binom{6}{6}.$$

3. Solve the recurrence equation

$$a_n - 4a_{n-2} = 1, \quad a_0 = 0, a_1 = 1.$$

4. You play a the following lottery with the casino. There is a hat on the table. A fair dice is thrown until the game ends.

   - If the dice shows one then $+1$ dollar is added to the hat and the game continues.
   - If the dice shows two then you immediately receive $+2$ dollars and the game continues.
   - If the dice shows three then $+3$ dollars are added to the hat and the game continues.
   - If the dice shows four then your receive half of the sum in the hat and the game ends.
   - If the dice shows five then the sum in the hat goes back to the casino and the game continues.
   - If the dice shows six then you receive the content of the hat and the game ends.

Let e_n be your total expected payoff if there are n dollars in the hat now.

Find the recurrence equation for e_n. You don't need to solve it.

5. Consider the recurrence equation

$$a_n - 5a_{n-1} + 3a_{n-2} = 0 \text{ for } n \geq 2, \quad a_0 = 1, a_1 = 2.$$

Find the generating function $g(t)$ for the sequence $(a_0, a_1, a_2, \ldots)$.

6. The sequence $(b_0, b_1, b_2, \dots)$ has a generating function

$$g(t) = \frac{3}{1-3t} + \frac{2t}{1-2t}.$$

Find b_0, b_1 and b_n.

9 Day 9. Generating functions in probability

Exercise 9.1. I throw a fair coin until the sequence HTH appears.

a) Draw the directed graph for this game.

b) Setup a system of equations for the sequences with the given endpoint and starting point at the start of the game.

c) Setup a system of equations for the sequences with the given starting point and endpoint at the end of the game.

d) Setup a system of equations for the non-terminal sequences with the given starting point.

Let N be the number of tosses, N_H — the number of H and N_T — the number of T till the end of the game.

e) Find expected values $\mathbb{E}(N)$, $\mathbb{E}(N_H)$, $\mathbb{E}(N_T)$.

f) Find probabilities $\mathbb{P}(N = 12)$, $\mathbb{P}(N_H = 13)$, $\mathbb{P}(N_T = 14)$.

g) Find variances $\mathbb{V}\text{ar}(N)$, $\mathbb{V}\text{ar}(N_H)$, $\mathbb{V}\text{ar}(N_T)$.

h) Find correlation $\mathbb{C}\text{orr}(N_H, N_T)$.

Note: you may use sympy from python or other software for symbolic computations.

Exercise 9.2. Alice and Bob throw a fair coin until the sequence $A = HTH$ or the sequence $B = HHT$ appears. Alice wins if the sequence A appears first, Bob wins if the sequence B appears first.

a) Draw the directed graph for this game.

b) Setup a system of equations for the sequences with the given endpoint and starting point at the start of the game.

c) Setup a system of equations for the sequences with the given starting point and endpoint at the end of the game.

d) Setup a system of equations for the non-terminal sequences with the given starting point.

Let N be the number of tosses, N_H — the number of H and N_T — the number of T till the end of the game.

e) Find expected values $\mathbb{E}(N)$, $\mathbb{E}(N_H)$, $\mathbb{E}(N_T)$.

f) What is the probability that Alice wins?

g) Find probabilities $\mathbb{P}(N = 12)$, $\mathbb{P}(N_H = 13)$, $\mathbb{P}(N_T = 14)$.

h) Find variances $\mathbb{V}\text{ar}(N)$, $\mathbb{V}\text{ar}(N_H)$, $\mathbb{V}\text{ar}(N_T)$.

i) Find correlation $\mathbb{C}\text{orr}(N_H, N_T)$.

j) Find conditional probabilities $\mathbb{P}(N = 12 \mid \text{Alice wins})$, $\mathbb{P}(N_H = 13 \mid \text{Alice wins})$, $\mathbb{P}(N_T = 14 \mid \text{Alice wins})$.

k) Find conditional expected values $\mathbb{E}(N \mid \text{Alice wins})$, $\mathbb{E}(N_H \mid \text{Alice wins})$, $\mathbb{E}(N_T \mid \text{Alice wins})$.

Note: you may use sympy from python or other software for symbolic computations.

Exercise 9.3.

Exercise 9.4.

Exercise 9.5.

Home assignment: day 9 $\to$ day 12

You can use sympy or any other software for symbolic computations.

1. I throw a fair coin until the sequence $HTHT$ appears.

   a) Draw the directed graph for this game.
   b) Setup a system of equations for the sequences with given endpoint and starting point at the start of the game.
   c) Setup a system of equations for the sequences with given starting point and endpoint at the end of the game.

   Let N be the number of tosses, N_H — the number of H and N_T — the number of T till the end of the game.

   d) Find expected values $\mathbb{E}(N)$, $\mathbb{E}(N_H)$, $\mathbb{E}(N_T)$.
   e) Find probabilities $\mathbb{P}(N = 12)$, $\mathbb{P}(N_H = 13)$, $\mathbb{P}(N_T = 14)$.

2. Alice and Bob throw a fair coin until the sequence $A = HHH$ or the sequence $B = THH$ appears. Alice wins if the sequence A appears first, Bob wins if the sequence B appears first.

   a) Draw the directed graph for this game.
   b) Setup a system of equations for the sequences with the given endpoint and starting point at the start of the game.
   c) Setup a system of equations for the sequences with the given starting point and endpoint at the end of the game.

   Let N be the number of tosses, N_H — the number of H and N_T — the number of T till the end of the game.

   d) Find expected values $\mathbb{E}(N)$, $\mathbb{E}(N_H)$, $\mathbb{E}(N_T)$.
   e) What is the probability that Alice wins?
   f) Find probabilities $\mathbb{P}(N = 12)$, $\mathbb{P}(N_H = 13)$, $\mathbb{P}(N_T = 14)$.

Demo for the quiz on day 10

1. I throw a fair coin until the sequence $HTHT$ appears. Draw the directed graph for this game.

2. Alice and Bob throw a fair coin until the sequence $A = HHH$ or the sequence $B = THH$ appears. Alice wins if the sequence A appears first, Bob wins if the sequence B appears first. Draw the directed graph for this game.

10 Day 10. Polya-Redfield counting

Definition 10.1. Necklace is a cirle of n coloured beads that are considered equivalent under rotations.
Cyclic group C_n is a collection of all possible rotations of a necklace.

Definition 10.2. Bracelet is a cirle of n coloured beads that are considered equivalent under rotations and flips (turn-overs).
Dihedral group D_n is a collection of all possible rotations and flips of a bracelet.

We will work with a set A with $\text{size}(A) = n$. The number of colours will be denoted as k. We will denote the set of all colorings of A by X. The group G acts on A and therefore on X by permuting the elements.

Definition 10.3. A group of actions G should satisfie two properties:

1. For every action g there should exist an anti-action denoted by g^{-1}.

2. For every two actions a and b there should exist the combined action $b \cdot a$. Here a is applied first, and b is applied second. Convention is similar to $\cos\sin(x)$ where sin-function is applied first and then cos-function is applied.

From these two principles it follows that every group G contains a «chill» action, identity element, denoted by e.

Definition 10.4. Consider an action $g = (12)(3)(4)(56)(789)$. The cyclic monomial of the action g is given by

$$\text{CM}_g(x_1, x_2, \ldots, x_n) = x_2 \cdot x_1 \cdot x_1 \cdot x_2 \cdot x_3 = x_1^2 x_2^2 x_3.$$

In general the cyclic monomial of an action is the product

$$\text{CM}_g(x_1, x_2, \ldots, x_n) = \prod_{i=1}^{n} x_i^{r_i}.$$

where r_i is the number of cycles of lenght i in the action g.

Definition 10.5. The cyclic index of the group G is the polynomial equal to the average of cyclic monomials for every action

$$\text{CI}(x_1, x_2, \ldots, x_n) = \frac{1}{\text{size}(G)} \sum_{g \in G} \text{CM}_g(x_1, x_2, \ldots, x_n).$$

Theorem 10.6 (Polya-Redfield theorem)**.** The generating function for all colorings of the set A into three colors (a, b and c) different with respect to group G is given by

$$g(a, b, c) = \text{CI}(a + b + c, a^2 + b^2 + c^2, a^3 + b^3 + c^3, \ldots, a^n + b^n + c^n).$$

Theorem 10.7 (Burnside lemma)**.** The number of colorings of the set A into k colors different with respect to the group G is given by

$$\text{size}(X/G) = g(1, 1, \ldots, 1) = \text{CI}(k, k, k, \ldots, k) = \frac{1}{\text{size}(G)} \sum_{g \in G} \text{size}(\text{fix}(g, X)).$$

Exercise 10.1. Consider a necklace with 3 beads.

a) How many different necklaces of k colours are there?

b) List all the necklaces of 3 colors using a generating function.

Exercise 10.2. Consider a bracelet with 3 beads.

a) How many different necklaces of k colours are there?

b) List all the necklaces of 3 colors using a generating function.

Exercise 10.3. Consider a necklace with 4 beads.

a) How many different necklaces of k colours are there?

b) List all the necklaces of 3 colors using a generating function.

Note: you may use sympy or other software for symbolic computations.

Exercise 10.4. Consider a bracelet with 4 beads.

a) How many different necklaces of k colours are there?

b) List all the necklaces of 3 colors using a generating function.

Note: you may use sympy or other software for symbolic computations.

Exercise 10.5. We consider two cubic dices identical if one is the rotation of the other.

a) Explicitely describe the group of actions G.

b) In how many ways can we label the faces of a dice with numbers from 1 to 6 if we should use every number exactly once?

c) In how many ways can we label the faces of a dice with numbers from 1 to 3 if we should use every number exactly twice?

d) Describe all possible dices with face labels from 1 to 3 using a generating function.

Note: you may use sympy or other software for symbolic computations.

Exercise 10.6. We consider two cubic dices identical if one is the rotation of the other.

a) Explicitely describe the group of actions G.

b) In how many ways can we paint the vertices of a cube in no more than k colors?

c) In how many ways can we paint the vertices of a cube in exactly k colors?

d) Describe all possible vertex colouring with no more than 3 colours using a generating function.

Note: you may use sympy or other software for symbolic computations.

Exercise 10.7. State the Polya-Redfield theorem for 4 colors.

11 Day 11. More Polya-Redfield counting

Exercise 11.1. We consider two tetrahedrons identical if one is the rotation of the other.

a) Explicitely describe the group of actions G.

b) In how many ways can we colour the faces of a tetrahedron with k colors?

c) Describe all possible tetrahedrons with faces coloured in 3 colors using a generating function.

Note: you may use sympy or other software for symbolic computations.

Exercise 11.2. We consider two tetrahedrons identical if one is the rotation of the other.

a) Explicitely describe the group of actions G.

b) In how many ways can we gems of no more than k colors in the vertices of a tetrahedron?

c) Describe all possible tetrahedrons with gems of 3 colors in the vertices using a generating function.

Note: you may use sympy or other software for symbolic computations.

Exercise 11.3. Let's consider a graph on 4 vertices. Two graphs are called isomorphic if they can be transformed one into another by reenumeration of vertices.

a) Explicitely describe the group of actions G.

b) In how many ways can we colour the edges of the complete graph K_4 with no more than k colors?

c) In how many ways can we colour the edges of the complete graph K_4 using exactly k colors?

d) Describe all possible graphs with 4 vertices and edges painted into no more than 3 colours using a generating function.

e) How many non-isomorphic graphs with 4 vertices are there?

Note: you may use sympy or other software for symbolic computations.

Exercise 11.4. Let's consider a 3×3 mosaic of no more than k colors. We consider two mosaics the same if one can be obtained from the other using rotations or flips.

a) Explicitely describe the group of actions G acting on mosaics.

b) How many mosaics are there?

c) Describe all possible mosaics with no more than 3 colours using a generating function.

Note: you may use sympy or other software for symbolic computations.

Exercise 11.5. Count the number of final positions in 3×3 noughts-and-crosses (tic-tac-toe) game.

Home assignment: day 11 $\to$ day 14

1. Let's consider a 3×3 mosaic of no more than k colors. We consider two mosaics the same if one can be obtained from the other using rotations or flips.

    a) Explicitely describe the group of actions G acting on mosaics.

    b) How many different mosaics are there?

    c) Describe all possible mosaics with no more than 3 colours using a generating function.

    Note: you may use sympy or other software for symbolic computations.

2. Watch the Mirek Olšák video «Burnside's lemma: counting up to symmetries» [Olš20].

    https://www.youtube.com/watch?v=D0d9bYZ_qDY, .

3. Watch the Thom de Boer video «Burnside's lemma: An Introduction to Group Theory» [Boe23].

    https://www.youtube.com/watch?v=Exot1TqjElY, .

Demo for the quiz on day 12

Let's consider a 4×4 mosaic of gems where each gem can be one of k types. Consider a $90°$ clockwise rotation of the mosaic.

1. Represent this rotation as a sequence of cycles and write the corresponging cycle monomial.

2. How many 4×4 mosaics are fixed by this rotation?

12 Day 12. Planarity and Euler's formula

Theorem 12.1 (Euler characteristic)**.** If a polyhedra with v vertices, f faces and e edges is drawn on a surface with h holes then

$$v - e + f = 2 - 2h.$$

Note: the hole should be a part of the polyhedra!

Definition 12.2. Complete graph K_n consists of n vertices. All edges are present.

Definition 12.3. Complete multipartite graph $K_{a,b,c,\ldots}$ consists of groups of nodes. The first group has a vertices, the second one — b vertices, the third one — c vertices, and so on. There are no edges between vertices of the same group. All edges between members of different groups are present.

Definition 12.4. The graph B is called a subdivision of a graph A if graph B is obtained from graph A by adding vertices on already existing edges.

Theorem 12.5 (Kuratowski's theorem)**.** A graph is planar if and only if it contains no subdivision of $K_{3,3}$ nor K_5.

Exercise 12.1. Prove the Euler characteristic formula. If a polyhedra with v vertices, f faces and e edges is drawn on a surface with h holes then

$$v - e + f = 2 - 2h.$$

Exercise 12.2. Consider a convex polyhedra with 20 vertices and 10 faces.

How many edges does it have?

Exercise 12.3. Consider a plane connected graph with v vertices and e edges that divides the plane into f regions (faces).

a) Find the value of the Euler characteristic $v - e + f$.

b) How the answer will change if the graph has c components?

Exercise 12.4. Consider a football made of p pentagons and h hexagons with three faces meeting at each vertex.

a) How many faces are there?

b) How many vertices are there?

c) How many edges are there?

d) How many pentagons should be on a football?

Exercise 12.5. Draw K_5, $K_{3,3}$ and $K_{2,2,2}$.

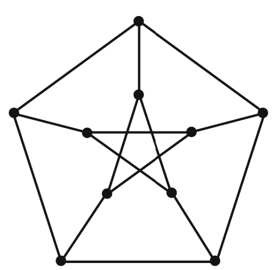
Draw any subdivision of K_5, $K_{3,3}$ and $K_{2,2,2}$.

Exercise 12.6. Check whether K_5 is planar.

Exercise 12.7. Check whether $K_{3,3}$ is planar.

Exercise 12.8. Can you draw $K_{3,3}$ on a cup with a handle without intersecting edges?

Exercise 12.9. Is the Petersen graph planar?



Exercise 12.10. a) Which complete graphs K_n are planar?

b) Which complete multipartite graphs $K_{a,b}$ are planar?

c) Which complete multipartite graphs $K_{a,b,c}$ are planar?

d) Which complete multipartite graphs $K_{a,b,c,d}$ are planar?

e) Which complete multipartite graphs $K_{a,b,c,d,e}$ are planar?

Exercise 12.11. Draw tetrahedron, cube, octahedron, dodecahedron, icosahedron.

You can play with models at https://polyhedra.tessera.li/, .

Exercise 12.12. Consider a convex polyhedra, non-necessary regular. Let k be the smallest number of edges of a face or the smallest cycle in the language of graph theory.

a) What is bigger $2e$ or $3v$?

b) What is bigger k or 6?

c) What are the possible values of k?

Exercise 12.13. Draw tetrahedron, cube, octahedron, dodecahedron, icosahedron.

Exercise 12.14. Consider a regular polyhedra where each face if a k-gon and degree of every vertex is d.

a What is bigger $2e$ or dv?

b What is bigger $2e$ or ke?

c Write the Euler characteristic identity in terms of d, k and e.

d What are the possible combinations of d and k?

13 Day 13. Graph colourings

Definition 13.1. By k-colouring we mean a colouring of vertices of a graph into no more than k given colours. A proper colouring assigns different colours do adjacent vertices. If a proper colouring of a graph exists then the graph is called k-colourable.

Definition 13.2. The chromatic number of a graph, denoted by $\chi(G)$, is the minimal number of colours k such that the graph G is k-colourable.

Exercise 13.1. By C_n we denote a cycle graph with n vertices.

1. Is the cycle C_5 colourable in 2 colours?
2. Is the cycle C_5 colourable in 3 colours?
3. Find the chromatic number $\chi(C_n)$ for $n \geq 2$.

Exercise 13.2. By P_n we denote a path graph with n vertices.

a) Find chromatic numbers $\chi(K_{50})$, $\chi(K_{20,30,40})$, $\chi(P_{255})$.

b) Find the general formulas $\chi(K_n)$, $\chi(K_{a,b,c})$, $\chi(P_n)$ for $a, b, c, n \geq 1$.

Exercise 13.3. You know that the chromatic number of a graph G is equal to 2026. You add one new vertex and some new edges, connecting the new vertex to the old ones.

What are the possible values of the new chromatic number?

Exercise 13.4. A connected graph contains 2026 edges.

What is the minimal possible chromatic number?

Exercise 13.5. You need to organize meetings of the following 7 groups of people: $G_1 = \{A, B, C\}$, $G_2 = \{C, D, E\}$, $G_3 = \{D, F\}$, $G_4 = \{A, G\}$, $G_5 = \{E, H\}$, $G_6 = \{E, B, G\}$, $G_7 = \{C, F, H\}$.

Yeach group should have a special meeting. Meetings may go parallel in time.

What is the lowest number of time slots needed?

Definition 13.3. The chromatic polynomial $\operatorname{chrom}(k, G)$ is the number of k-colourings of a graph G. We consider two colourings different if they differ in at least one vertex.

Exercise 13.6. Find some chromatic polynomials!

a) Find $\operatorname{chrom}(k, K_n)$ for $n \geq 1$.

b) Find $\operatorname{chrom}(k, C_n)$ for $n \geq 2$.

c) Find $\operatorname{chrom}(k, G)$ for the empty graph G with no edges and n vertices.

d) Find $\operatorname{chrom}(k, T_n)$ for a tree with n vertices.

Exercise 13.7. The chromatic number of some graph is $\chi(G) = 4$. What can be said about the values $\operatorname{chrom}(1, G)$, $\operatorname{chrom}(2, G)$, $\operatorname{chrom}(3, G)$ and $\operatorname{chrom}(4, G)$?

Theorem 13.4 (Birkhoff and Lewis)**.** Notation:

- $G - e$ is the graph G with the edge e removed;
- G/e is the graph where the ends of the edge e are glued into one vertex and duplicated edges removed.

$$\operatorname{chrom}(k, G) = \operatorname{chrom}(k, G - e) - \operatorname{chrom}(k, G/e).$$

Exercise 13.8. Prove the Birkhoff and Lewis formula

$$\operatorname{chrom}(k, G) = \operatorname{chrom}(k, G - e) - \operatorname{chrom}(k, G/e).$$

Exercise 13.9. Find some more chromatic polynomials!

a) Find $\operatorname{chrom}(k, K_4 - e)$ for $n \geq 1$.

b) Find $\operatorname{chrom}(k, K_{1,3})$ and $\operatorname{chrom}(k, K_{1,5})$.

Exercise 13.10. You know that $\operatorname{chrom}(k, G) = k(k-1)^{n-1}$. What can you conlude about the graph G?

Exercise 13.11. Some properties of the chromatic polynomial.

a) What is the degree of the polynomial $\operatorname{chrom}(k, G)$ if the graph G has n vertices?

b) What is the value of the constant term in $\operatorname{chrom}(k, G)$?

c) What is the value of the coefficient of k^n in $\operatorname{chrom}(k, G)$?

d) What is the value of the coefficient of k^{n-1} in $\mathrm{chrom}(k, G)$ if the graph G has e edges?

Exercise 13.12. Is it possible that $\mathrm{chrom}(k, G) = k^4 - 3k^3 + 3k^2$?

Exercise 13.13. Imagine the graph G consists of two components, A and B. Each component is connected, but there are no edges between A and B.

a) How would you obtain $\mathrm{chrom}(k, G)$ from $\mathrm{chrom}(k, A)$ and $\mathrm{chrom}(k, B)$?

b) What is the value of the coefficient of k in $\mathrm{chrom}(k, G)$?

Definition 13.5. The Ramsey number $\mathrm{Ramsey}(G, H)$ is the minimal number of vertices n of a complete graph K_n such that any colouring of its edges into blue and red contains either a red copy of G or a blue copy of H.

The Ramsey number $\mathrm{Ramsey}(K_a, K_b)$ is simply denoted by $\mathrm{Ramsey}(a, b)$.

Exercise 13.14. Find $\mathrm{Ramsey}(1, b)$ and $\mathrm{Ramsey}(2, b)$.

Exercise 13.15. Find $\mathrm{Ramsey}(6, 6)$ and be the greatest mathematician of the century (or maybe millenium).

Exercise 13.16. Find some small Ramsey numbers:

a) Find $\mathrm{Ramsey}(P_2, P_2)$, $\mathrm{Ramsey}(P_2, P_3)$, $\mathrm{Ramsey}(P_3, P_3)$.

b) Find $\mathrm{Ramsey}(C_2, C_2)$, $\mathrm{Ramsey}(C_2, C_3)$, $\mathrm{Ramsey}(C_3, C_3)$, $\mathrm{Ramsey}(C_3, C_4)$, $\mathrm{Ramsey}(C_4, C_4)$.

Exercise 13.17. Watch Trefor Bazett's video «Why complete chaos is impossible: Ramsey theory», [Baz23]:

https://www.youtube.com/watch?v=ZykdsnIpXNk,

14 Day 14. How to avoid combinatorics?

Theorem 14.1 (Linearity of expected value)**.** The identity $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$ holds provided that both sides are defined.

Exercise 14.1. You are in a cave and you see 10 passages. The first one has lenght 1 km, the second one — 2 km, the third one — 3 km and so on. Only the last passage is a way out, all others end with an impasse.

You wish to get out and try passages in a random order. If you try a bad passage you go till its end and then go back to the original position.

a) What is your expected walking distance if you mark already visited bad passages and do not visit them twice?

b) What is your expected walking distance if you are nervous and may visit already explored bad passage?

Exercise 14.2. Maria has 30 pairs of shoes. Her dog Charlie chewed 17 random shoes. Charlie is indifferent between left and right shoes, they are all delicious!

a) How many complete pairs on average are still available to Maria?

b) How many pairs are completly chewed?

Exercise 14.3. In my pocket I have 25 baht in coins. There is exactly one 5 baht coin among them. I take out the coins at random order one by one. I stop when I pull out the 5 baht coin.

a) What is the average value of taken out coins?

b) How the answer will change if there are exactly two 5 baht coins in my pocket?

Exercise 14.4. Maria has 100 bracelets. During one year every morning she picks one of them with equal probabilities to wear. She can pick the same bracelet on different days.

a) How many bracelets on average she will never wear?

b) How many bracelets on average she will wear two times or more?

Exercise 14.5. I have n students and I remember that all of them got different grades for the exam. As a true probability teacher I enter the grades of students in a completly random order.

What is the expected number of students that will see the correct grade?

Exercise 14.6. There are 20 types of desserts in a café. In the queue there are 10 clients. Each client prefers one of the desserts at random with equal probabilities.

a) What is the average number of dessert types bought?

b) What is the average number of dessert types bought if each client will not buy any dessert with probability 0.3?

Exercise 14.7. There are 27 kids performing round dance. We say that the kid is tall if he is taller than two his neighbors.

What is the average number of tall kids?

Exercise 14.8. We put randomly 7 pens in 10 boxes one by one choosing each box with equal probabilities.

a) What is the average number of empty boxes?

b) What is the average number of boxes with more than one pen?

Exercise 14.9. Inside a cereal box there is a toy of one of 30 types. All types have equal probability.

If you collect all the types of toys you will get a big prize.

How many cereal boxes should you buy on average to collect all the types of toys?

Exercise 14.10. You draw randomly 10 cards from the standard deck of 52 cards.

How many aces are drawn on average?

Exercise 14.11. You drop needle of length 1 on a grid of parallel lines with the distance 1 between lines.

What is the probability that the needle will cross one of the parallel lines?

Exercise 14.12. Consider a random graph on $n \geq 5$ vertices where each edge is drawn independently with probability p.

a) What is the average number of isolated vertices?

b) What is the average number of edges?

c) What is the average number of K_4 subgraphs?

d) What is the average number of C_3 subgraphs?

e) What is the average number of subgraphs that are trees with 5 vertices?

15 Day 15. Absolute happiness!

Final demo

This is a demo version of the final exam. Final will consist of 6 problems: generating functions in probability, Polya-Redfield counting, planarity and Euler formula, graph colouring, linearity of expected value and one old topic. Some problems will be similar to demo

You may use one A4 cheat sheet in any form handwritten, printed, aquarelle painting...

1. [old topic] Write the Taylor expansion at $x_0 = 0$ of the function

$$f(x) = (1 - 5x^2)^{1/2}.$$

2. [generating functions in probability] I throw a coin until the sequence HTT appears. The coin is not fair and shows H with probability 0.4.

   a) Draw the directed graph for this game.

   b) Setup a system of equations for the sequences with given endpoint and starting point at the start of the game.

   c) Explain how would you calculate expected number of coin tosses from the start till the end of the game.

      Note: you don't need to do actual calculations nor to solve the system, just describe how can you extract the expected value using derivatives.

3. [Polya-Redfield counting] Consider a 2×2 mosaic of gems, where each gem may be of one of k possible types. We consider mosaics identical if the can be transformed one into another by rotation or flip.

   a) List all elements of the group of actions that act on mosaics.

   b) Write each action as a combination of cycles and provide its cyclic monomial.

   c) Find the cyclic index and the number of different possible colourings.

4. [planarity and Euler formula] The convex polyhedron consists only of triangular and hexagonal faces with three faces meeting at each vertex.

   How many triangular faces are there?

5. [graph colouring] Consider the graph with two components. The first component is a K_4 graph, the second one — C_4 graph. Components are non connected between them. You have k colours available.

   In how many ways can you color the vertices of this graph if adjacent colours can't be the same?

6. [linearity of expected value] There are 20 types of desserts in a café. In the queue there are 10 clients. Each client prefers one of the desserts at random with equal probabilities.

   a) What is the average number of dessert types bought?

   b) What is the average number of dessert types bought if each client will not buy any dessert with probability 0.3?

Final exam

1. [generating functions in probability] I throw a fair coin until the sequence HHH appears.

   a) Draw the directed graph for this game.

   b) Setup a system of equations for the sequences with given endpoint and starting point at the start of the game.

   c) Explain how would you calculate the expected number of tails from the start till the end of the game. Note: you don't need to do actual calculations nor to solve the system, just describe how can you extract the expected value using derivatives.

2. [Polya-Redfield counting] Consider a 3×3 mosaic of gems, where each gem may be of one of k possible types. We consider mosaics identical if the can be transformed one into another by rotation. Flips are not available!

   a) List all elements of the group of actions that act on mosaics.

   b) Write each action as a combination of cycles and provide its cyclic monomial.

   c) Find the cyclic index and the number of different possible mosaics for $k = 10$ gem types available.

3. [planarity and Euler formula] Rhombicosidodecahedron has 20 triangular faces, 30 square faces and some number of pentagonal faces. The total number of vertices is 60.

   a) How many pentagonal faces does rhombicosidodecahedron have?

   b) How many edges are meeting at every vertex?

4. [graph colouring] Consider the graph with three components. The first component is a C_4 graph, the second one — a tree with 10 vertices, the third one — an isolated vertex. Components are non connected.

   a) Draw an example of such a graph.

   b) You have 10 colours available. In how many ways can you color the vertices of this graph if adjacent vertices should be painted differently?

5. [linearity of expected value] We put randomly 7 pencils in 10 boxes one by one choosing each box with equal probabilities.

   a) What is the average number of empty boxes?

   b) What is the average number of boxes with more than one pencil?

6. [old topic] The generating sequence $g(t)$ of an integer sequence $(c_0, c_1, c_2, \dots)$ satisfies the equation

$$g(t) = 1 + t \cdot g(t) \cdot g(t).$$

   (a) Solve the equation for $g(t)$.

   (b) Which of the two roots should we choose if all c_n are positive?

   (c) Find c_3.

16 Yet to discover

Inclusions and exclusions

Proofs to DIE for

Wandermonde's identity

Exercise 16.1. Simplify

$$\binom{a}{0}\binom{b}{m}+\binom{a}{1}\binom{b}{m-1}+\binom{a}{2}\binom{b}{m-2}+\cdots+\binom{a}{m}\binom{b}{0}.$$

$$\binom{a}{0}\binom{m}{0}+\binom{a}{1}\binom{m}{1}+\binom{a}{2}\binom{m}{3}+\cdots+\binom{a}{m}\binom{m}{m}.$$

$$\binom{a}{0}^2+\binom{a}{1}^2+\binom{a}{2}^2+\cdots+\binom{a}{a}^2.$$

Exercise 16.2. There are $n = 500$ fishes in the pond, $t = 30$ of them are tagged. You capture $c = 5$ fishes at random without replacement. Let the random variable Y be the number of captured tagged fishes. Find the probabilities $\mathbb{P}(Y = 0)$, $\mathbb{P}(Y = 1)$, $\mathbb{P}(Y = 2)$, $\mathbb{P}(Y = y)$.

Exercise 16.3. There are $n = 24$ candies in the jar, 14 with orange flavour and 10 with ananas flavour. Your take 6 candies at random. Let Y be the number of candies with orange flavour you picked. Find the probabilities $\mathbb{P}(Y = 0)$, $\mathbb{P}(Y = 1)$, $\mathbb{P}(Y = 2)$, $\mathbb{P}(Y = y)$.

Exercise 16.4. A bridge hand (random 13 cards out of 52-card standard deck) is dealt.

a) What is the probability that the hand contains no spades?

b) What is the probability that the hand contains exactly 5 hearts?

c) What is the probability that the hand contains at most two aces?

Exercise 16.5. Wandermonde-like identities in a funny notation. Here I intentionally omit the running index i, I write the first summation term and specify with $\uparrow$ and $\downarrow$ which terms goes up or down in the sum.

a) Simplify and rewrite in a standard notation

$$\sum \binom{n}{0\uparrow}\binom{n\downarrow}{k}.$$

b) Simplify and rewrite in a standard notation

$$\sum \binom{a\uparrow}{a}\binom{n\downarrow}{k}.$$

c) Simplify and rewrite in a standard notation

$$\sum \binom{b}{0\uparrow}\binom{n}{k\downarrow}.$$

Exercise 16.6. Simplify

$$\sum_{a+b+c=k}\binom{n_1}{a}\binom{n_2}{b}\binom{n_3}{c}.$$

17 Sources of Wisdom

[Ver21] Veritasium. Newton changed the game. 2021. URL: https://youtu.be/gMlf1ELvRzc?si=VASoVKQj3Pk-wstA. How Newton discovered binomial coefficients for non-integer values and used them to calculate pi, 19 minutes.

[Ver20] Veritasium. This equation will change how you see the world. 2020. URL: https://youtu.be/ovJcsL7vyrk?si=kh8wjM2e6dgLSXzj. How chaos may arise from a simple recurrence equation, 19 minutes.

[Olš20] Mirek Olšák. Burnside's lemma: counting up to symmetries. 2020. URL: https://www.youtube.com/watch?v=D0d9bYZ_qDY. Counting 3 by 3 mosaics with Burnside lemma, 13 minutes.

[Boe23] Thom de Boer. Burnside's lemma: An Introduction to Group Theory. 2023. URL: https://www.youtube.com/watch?v=Exot1TqjElY. Counting bracelets using Burnside lemma, author uses the term necklace instead of bracelet, 15 minutes.

[Baz23] Trefor Bazett. Why complete chaos is impossible: Ramsey theory. 2023. URL: https://www.youtube.com/watch?v=ZykdsnIpXNk. Classic examples of Ramsey theory: friends and strangers theorem, high dimensional tic-tac-toe, Van-der-Waerden theorem with idea of proof, 23 minutes.

[Gra94] Ronald L Graham. Concrete mathematics: a foundation for computer science. Pearson Education India, 1994. A classic captivating (really funny sometimes!) treatise on combinatorics from masters, 690 pages.

[Har+10] John J Harris et al. Combinatorics and graph theory. Springer-Verlag New York, 2010. Accessible book with a lot of pictures, stories and nice explanations, 355 pages.

[Gui25] David Guichard. An Introduction to Combinatorics and Graph Theory. 2025. URL: https://www.whitman.edu/mathematics/cgt_online/cgt.pdf. Short open access introductory book with pictures and good intuition, 155 pages.

[Sac21] SackVideo. Counting with Calculus: The Magic of Generating Functions. 2021. URL: https://www.youtube.com/watch?v=dLiT9axMDrg. Fibonacci numbers and graphs counted with generating functions, 22 minutes.

[3Bl22] 3Blue1Brown. Olympiad level counting. 2022. URL: https://www.youtube.com/watch?v=bOXCLR3Wric. Hard combinatorics problem solved with generating functions and complex numbers, 35 minutes.

[Hui21] Mollee Huisinga. Pólya's counting theory. 2021. URL: https://www.whitman.edu/Documents/Academics/Mathematics/Huisinga.pdf. Many examples for Polya enumeration theorem and Burnside theorem with pictures, 46 pages.

[Bax05] Nick Baxter. The Burnside Di-Lemma: Combinatorics and Puzzle Symmetry. 2005. URL: https://www.baxterweb.com/puzzles/burnside5.pdf. Examples for Burnside lemma with pictures from puzzle expert, chapter from the book «Tribute to a Mathemagician», 14 pages.

[Kem25] Kempu33334. Burnside lemma. 2025. URL: https://kempu33334.github.io/Handouts/Burnside-Lemma-Handout/main.pdf. Short intro to Burnside Lemma with exercises without proof (probably AI used), 7 pages.